Project Proposal

Course: Data, Data Storage, Data Collection — 16.09.2026

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Central question

How often does Vélib' actually fail you, and is everyone equally exposed?

A station with no bikes looks identical whether its equipment is broken or nobody is riding. Our project is built around telling those two cases apart, and around measuring failure from the user's point of view rather than the station's. Four steps, each answering an objection raised by the previous:

1. **When is a station unusable?** No bikes, no free docks, not renting, or a frozen state.
2. **Does it matter?Is there a usable station next door?** A dead station surrounded by others is an inconvenience; an isolated one is a lost trip, so we measure failure within the user's walking radius.
3. **When is it worst?** Weather, commuting rhythm, school holidays.
4. **Is it broken, or simply unused?** Independent measures of cycling activity separate equipment failure from absent demand.

The analysis will stay descriptive throughout episode reconstruction, aggregation, comparison across periods and areas, and we will consult you before introducing anything beyond that.

Data

Primary source, collected by us. Vélib' Métropole publishes station status through an open GBFS feed, updated continuously. It reports the present state only; no history is published anywhere. We archive the raw `station_status` feed every 5 minutes and the `station_information` reference hourly, until 16 November. **Collection will start on 20.09**, and a pilot of 5 snapshots over 1,519 stations has validated the pipeline. Expected volume: about 26 million observations, 1 GB of compressed raw JSON.

Supporting sources. Paris bike-counter data (opendata.paris.fr) measures cycling activity independently of Vélib', which is what lets us distinguish a broken station from a quiet neighbourhood. Open-Meteo supplies hourly weather without a key. The official school calendar marks the Toussaint holidays, 17 October to 2 November, a natural break in commuting demand falling inside our window. Coordinates are reverse-geocoded to communes via the Base Adresse Nationale.

Anticipated challenges

Feed latency. In our pilot, the gap between our capture time and a station's own `last_reported` was never below 20.8 minutes, median 30.8. We cannot study dynamics below roughly half an hour.

Unreliable metadata. In 46% of pilot rows, bikes and docks do not add up to the stated capacity, leaving about 1,256 unaccounted slots network-wide. Fourteen stations report a state frozen for over 24 hours, one of them by more than five years. Separating genuine failure from reporting artefact is our main cleaning problem.

Uneven coverage of our control. The bike counters are a Ville de Paris facility, while 305 of our 1,519 stations lie outside Paris. We will either restrict the demand analysis to the Paris subset or accept weaker inference in the suburbs, and will state which.

Continuous collection. Ten weeks of uninterrupted recording is an operational risk. We run the collector on two independent machines and log every HTTP attempt, so gaps are measurable rather than invisible.

Storage. The project mixes a large append-only time series, a slowly changing reference table, small external hourly series and a geographic layer; we will compare a relational schema against date-partitioned columnar files on size, ingestion time and query cost rather than assert a choice.